COVID-19 : Primary Thromboprophylaxis for Hospitalized Patient

Objective:

To outline a practical approach for the use of thromboprophylaxis in critically ill and non-critically ill patients hospitalized with COVID-19.

Background:

COVID-19 is associated with an increased risk of thrombosis. Observational studies early in the pandemic reported high rates of venous thromboembolism (VTE) in hospitalized patients with COVID-19, with pooled incidences approximating 10-30% despite routine thromboprophylaxis. These rates are substantially higher than those historically documented for medically ill hospitalized patients without COVID-19 (less than 2%).

Patients hospitalized with COVID-19 commonly carry traditional provoking factors for venous thromboembolism (VTE), including immobility, obesity, and advanced age. Further, SARS-CoV-2 infection directly inflames the endothelium, resulting in endotheliopathy and in-situ thrombosis, which may explain the disproportionately higher rates of PE than DVT in this patient population. Viral infection also drives a robust systemic inflammatory response which stimulates procoagulant pathways. Thrombogenicity in COVID-19 correlates with disease severity, which is reflected by higher VTE rates in patients needing ICU-level care. In turn, diagnosis of VTE is a poor prognostic marker in hospitalized patients with COVID-19.

In the absence of contraindications, all patients hospitalized with COVID-19 warrant pharmacologic thromboprophylaxis during their hospitalization. Multiple randomized control trials (RCTs) have evaluated the safety and efficacy of primary thromboprophylaxis at higher than standard dosing in selected critically ill and non-critically ill hospitalized patients with COVID-19 at low risk of bleeding. The purpose of this document is to provide recommendations for dosing and duration of pharmacologic thromboprophylaxis in patients hospitalized with COVID-19, on the basis of the results from these trials.

Thromboprophylaxis for non-critically ill patients:

Summary of the evidence: Three RCTs compared therapeutic-dose to low- or intermediate-dose low molecular weight heparin (LMWH) and unfractionated heparin (UFH) in hospitalized non-critically ill patients with COVID-19 requiring low flow supplemental oxygen. Patients with high bleed risk were excluded. Therapeutic-dose LMWH/UFH improved clinical outcomes, prevented progression to critical illness, reduced rates of major thromboembolism, and trended to reduce all-cause mortality according to a meta-analysis of the three trials. This benefit seems to be specific to LMWH/UFH and is not seen with DOACs. When compared to standard thromboprophylaxis, Rivaroxaban 20 mg daily for 30 days did not improve clinical outcomes and increased risk of bleeding in one RCT. Details of each trial are summarized in **Table 1**.

Anticoagulant dose and timing: Non-critically ill patients hospitalized for COVID-19 at low risk for bleeding should receive primary thromboprophylaxis at therapeutic dosing rather than standard or intermediate dosing. Therapeutic-dose thromboprophylaxis should be initiated upon admission, and continued throughout hospitalization for up to 14 days, in the absence of bleeding complications. Suggested regimens and dosing schedules are listed in **Table 5**. Patients with risk factors for bleeding should remain on standard-dose thromboprophylaxis throughout admission (see **Table 3** for variables associated with increased bleeding risk).

Anticoagulant choice: LMWH or UFH should be used for thromboprophylaxis. We prefer LM
former carries a lower risk of bleeding, heparin-induced thrombocytopenia, and minimizes pers
LMWH formulation listed in **Table 5** is appropriate to use. Direct oral anticoagulants (DOACs)
results and should not be used for thromboprophylaxis in this setting.

Thromboprophylaxis for critically ill patients:

For the purposes of this document, COVID-19-related critical illness is defined as the need for cardio-pulmonary organ support, such as oxygen by high-flow nasal canula, mechanical or non-invasive positive pressure ventilation, vasopressors, inotropes, or extra-corporeal membrane oxygenation (ECMO).

Summary of the evidence: In critically ill patients with COVID-19, two RCTs compared therapeutic to standard-dose thromboprophylaxis with LMWH/UFH, and two RCTs compared intermediate to standard-dose thromboprophylaxis. Compared to controls, neither therapeutic nor intermediate dose LMWH/UFH improved clinical outcomes or mortality. Further, patients who received therapeutic dose anticoagulation trended to have higher rates of bleeding. Details of each trial are summarized in **Table 2**.

Anticoagulant regimen: In patients with critical illness, thromboprophylaxis should be administered in the form of LMWH or UFH at standard dosing (see **Table 5**). UFH should be reserved for patients with contraindications to LMWH such as renal failure. Thromboprophylaxis should be initiated upon admission and continued throughout the duration of hospitalization.

Post-discharge thromboprophylaxis:

Extended post-discharge thromboprophylaxis for up to 35 days with low-dose rivaroxaban 10 mg daily may be considered in select patients with high VTE risk (see **Table 4**), no risk factors for bleeding (see **Table 3**), and not receiving antithrombotic therapy for other indications. This recommendation is based on one RCT (n=320) in which rivaroxaban 10 mg daily for 35 days reduced rates of VTE without increasing bleeding in patients hospitalized for COVID-19. Notably, only patients who had a high thrombotic risk and a low risk of bleeding were selected for this study.

Table 1. RCTs of thromboprophylaxis in non-critically ill patients with COVID-19

CRNMB= clinically relevant non-major bleed

[*]In the main cohort of both critically-ill and non-critically ill groups, rates of venous and arterial thromboembolism were reduced in the treatment arm compared to controls (10.9% vs. 29.0%, $p<0.001$)

[**]Defined as any VTE, myocardial infarction, stroke, systemic embolism, and major adverse limb event.

Table 2. RCTs of thromboprophylaxis in critically ill patients with COVID-19

*In the main cohort of both critically-ill and non-critically ill groups, rates of venous and arterial thromboembolism were reduced in the treatment arm compared to controls (10.9% vs. 29.0%, $p<0.001$)

Table 3. Risk factors for bleeding

Table 4. Risk factors for VTE in hospitalized patients

Table 5. Options for thromboprophylaxis in patients hospitalized with COVID-19.

*For therapeutic dosing at extremes of weight, consider consulting a thrombosis expert.

**For standard thromboprophylaxis dosing at extremes of weight (<40kg or >100kg), refer to the Thrombosis Canada Guidelines “Thromboprophylaxis: Hospitalized Medical Patients”.

¶Creatinine clearance, calculated by Cockcroft-Gault formula.

¶¶Limit use of UFH to patients with contraindications to LMWH (allergy or CrCl <20ml/min).

Special considerations:

Patients who deteriorate during their hospitalization: There is limited evidence to guide thromboprophylaxis management of hospitalized patients who eventually develop critical illness during their admission. Nonetheless, bleeding risk is substantially increased in critically ill patients with COVID-19 receiving therapeutic-dose anticoagulation. As such, hospitalized patients receiving therapeutic-dose anticoagulation should be stepped down to standard-dose anticoagulation if they deteriorate to require organ support, as already defined.

Management of thromboprophylaxis in patients with renal failure: LMWH agents can bioaccumulate in patients with renal impairment. Therapeutic-dose enoxaparin and dalteparin should be avoided in patients with creatinine clearance (CrCl) less than 30 ml/min, as calculated by the Cockcroft-Gault equation. Therapeutic-dose tinzaparin may be considered until CrCl of 20ml/min.

For patients admitted with COVID-19 and CrCl less than 20 ml/min, including those requiring dialysis, unfractionated heparin (UFH) or dalteparin (at standard prophylactic dose) could be considered for thromboprophylaxis. Non-critically ill patients selected to receive therapeutic-dose thromboprophylaxis should receive IV UFH to target a therapeutic aPTT or anti-Xa level per institutional protocol. Critically ill patients with severe renal dysfunction could receive standard prophylactic dose subcutaneous UFH (5000U SQ BID or TID) or dalteparin 5000U SQ daily.

Patients receiving antithrombotic therapies for other indications at the time of admission

a) Patients receiving antiplatelet therapy: Hospitalized patients taking dual antiplatelet therapy (DAPT) should receive standard-dose rather than therapeutic-dose thromboprophylaxis in hospital, as bleed risk is increased. For select patients on single antiplatelet therapy in whom thrombotic risk significantly outweighs bleed risk, addition of therapeutic-dose thromboprophylaxis can be considered on a case-by-case basis. Consultation with a thrombosis or hematology expert may be appropriate in this setting.

b) Patients receiving therapeutic-dose anticoagulation: Patients already on therapeutic-dose a
other indications at the time of admission should remain on therapeutic-dose anticoagulation i
appropriate, irrespective of the severity of illness. Patients receiving DOACs or VKAs prehospi
transitioned to LMWH while acutely ill. Oral anticoagulation can be resumed when the patient is nearing discharge and able to tolerate oral intake.

Use of biomarkers such as D-dimer to guide thromboprophylaxis: There is no evidence to support the use of clinical biomarkers to guide thromboprophylaxis dosing or duration in patients with COVID-19. Although an elevated D-dimer may be associated with increased thrombotic risk, the benefit of therapeutic-dose thromboprophylaxis in this patient population is observed irrespective of the baseline D-dimer level. As such, neither the D-dimer nor other biomarkers should be incorporated into thromboprophylaxis decisions.

Limitations to this guide:

The phenotype of SARS-CoV2 illness is rapidly evolving. The studies used to formulate the above recommendations recruited patients during early stages of the pandemic, prior to the advent of universal vaccination and the emergence of novel strains. Further, these studies selected for a patient subpopulation at low risk of bleeding. As such, patient care should be individualized, and thromboprophylaxis decisions should be approached on a case-by-case basis. When possible, the patient's preferences should be incorporated into treatment decisions, and shared decision-making should be carried out between members of the care team.

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